Talula Wilmot
Writing 105
Professor Kenny Smith

Could we be witnessing the end of Giants? The Stressors Threatening Some of the World's Oldest Organisms

Introduction:

Imagine that you are a young and eager graduate student. At 30 years old you are attending the university of North Carolina hoping to study climate change and glaciology using tree cores in Great Basin national park. You have spent the better part of your adult life in the academic world and you are hoping to make a difference in the face of new findings about the risk that climate change poses to the human race. While it is the early 1960's and you don't yet have the vast conception of anthropogenic damage to the planet you are beginning to see the effects of emissions in the face of the industrial boom that followed the great depression. This was Donald Rusk Currey. He was a dendrochronologist studying ice age glaciology using bristlecone pine to age the glacial features that the trees grow on top of. This forest housed trees over 4,000 years old making them highly valuable research subjects as their rings held climatic data over their entire life period. After receiving permission from the US forest service he took the core from a very large bristlecone pine. These trees are some of the oldest on the planet. They grow at high elevations in extreme conditions from lighting, to drought, extreme temperatures and wind. They have endured thousands of years of adversity but it wasn't until Donald Currey that they had met their human match. After the first core was taken the course of events became a bit blurry.

Some accounts say that the tree core got lodged and stuck inside the tree. Some say that Currey didn't know how to core a tree of such a large size. Some say that Currey requested a full cross section to better examine the rings. One way or another he was granted rightful permission to cut the tree down and after doing so he revealed to himself that the tree known as Prometheus was over 4,900 years old making it the oldest known tree to date. The stump of Prometheus still remains in the grove on Wheeler Peak[1]. One can only imagine the devastation felt by Currey after what he had done. What would ensue were a series of interviews, none able to create a clear narrative, and ultimately the forest service would come to implement stricter policy on the ability for researchers to access trees in ancient groves.

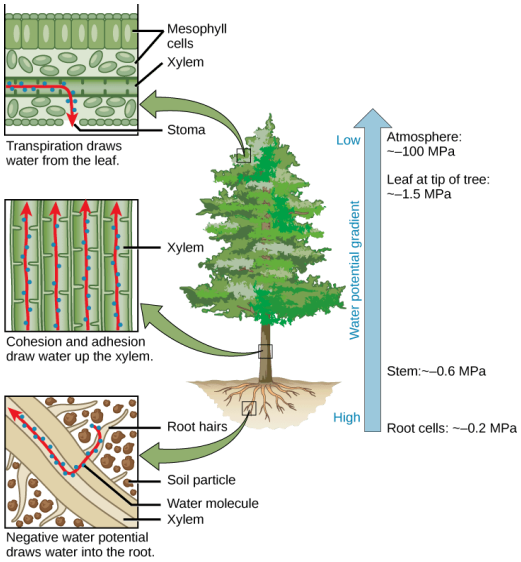
At the time of incidence this caused a great deal of uproar within the scientific community. This event would be a reminder of anthropogenic threats to ancient forests. While the oldest of its time in 2012 another bristlecone pine proved to be 5,065 years old taking the top spot from Prometheus. On an individual scale the death of this tree while a great tragedy provided immense insight on millenia of climatic changes. On a mass scale there would be much greater threats to the health of these forests though ultimately still anthropogenic in nature. Giant Sequoias are one such tree within these forests that are of great interest to the scientific community. They are a well known fire adapted tree with thick

bark to act as an insulator, and serotinous cones, meaning that they only drop their seeds after a fire to reduce competition. Despite these adaptations, the frequency of fire as well as its severity poses a huge threat to the species. This is largely due to the 20th century push for fire suppression following the idea that all fire was harmful to life. When we think of fire suppression we can think of the mid 20th century enactment of smokey the bear. “Only you can stop wildfire” was his signature phrase. Though true, today humans are the leading cause of wildfire, it presented the idea that all fire was harmful to the environment. This created an unnatural build up of organic fuel which has created increasingly intense and damaging fires in recent years. Despite these findings the direct rate of fire on Giant Sequoia mortality is still relatively unknown as these high severity burns are relatively modern, and delayed mortality in the years post fire is not well captured[2].

We can then pose the question, how much fire is too much? It is not simply an effect of large fires causing tree mortality, as it was previously part of the natural landscape for millions of years. These adverse effects are both a factor of multiple fires leaving burn scars making trees more vulnerable to proceeding burns as well as compound effects of drought. During the historic 2012-2016 drought that was the most severe in at least 120 years, sequoias, while showing a very low mortality, did demonstrate signs of extreme stress[3]. The stress of drought also makes several coniferous species vulnerable to disease, most notably bark beetles. Stress ultimately leads to greater vulnerability making a once useful fire detrimental.

There are several measures of stress that can be used to assess the state of a tree from the leaf all the way to a community level. One of the most effective strategies for measuring stress at the leaf level is through water potential measurements. This can be used in combination with remote sensing to assess whole canopy water content (CWC) and foliage die back through simple observation. Collectively these provide a clear picture of the level of stress trees are under which can create vulnerability as well as mortality estimates.

Water Potential Measurements and Remote Sensing:



Giant sequoias currently have an incredibly limited distribution with the majority of groves residing in Sequoia Kings Canyon National Park. This is due to historic climatic changes such as glaciation as well as specialized soil moisture needs and low resilience of saplings to drought. Recruitment is also low in the last century as fire suppression prevents the release of seeds from sequoia cones. This makes them key species of interest as despite this small distribution across the western Sierra Nevada they maintain incredible resilience to harsh environmental conditions with especially long life

spans. Though saplings typically do not survive long periods of drought mature trees have developed alternative tactics. When thinking about the giant sequoias ability to uptake water you can think of it like a straw. It is constantly sipping water via the negative water potential gradient from root to atmosphere. If you are a large tree like a Sequoia this water comes from your extensive net like root system that both prevents you from tipping over but more importantly increases your surface area within the soil and allows you to uptake water more efficiently. In the winter when rains are heavy and water is abundant you typically do not have to suck too hard to get water. In the summer, late august, when you are in your fifth month of drought, supplies are a bit scarcer. As we mentioned earlier, since these roots are used to hold you up they are extended far and wide, much different from a phreatophyte that would have one very long tap root extending directly down into the water table, a special adaptation for arid climates. When the water availability decreases the strength you need to suck water up increases. This principle is generally known as transpiration. A plant will take water up from its roots, disperse it to its limbs and the stomata will release the water on the leaf's surface so it may evaporate. Now back to late august. You have begun to suck water up through your straw as hard as you possibly can but the straw now longer has a steady stream. Instead you have reached the end of your cup and the final inhales are filled with pockets of water and air. Once these pockets of air enter your xylem your tissue becomes vacuum locked with the air bubble and essentially dies off as it is no longer able to transpire water.

One key way to measure the extent of this result is through water potential measurements of the leaves. A water potential measurement typically begins by accessing leaves from the upper canopy of the tree where sun exposure is constant throughout the day. This can be done by climbing trees or using large pruners to clip from the periphery. These samples are collected at predawn, before the stomata have opened and midday when they are in full use. Once collected they are quickly covered and bagged to prevent further photosynthesis and brought back to be measured. The measurement involves what is known as a pressure bomb. The leaf is inserted into the pressure bomb where gas fills the chamber pushing on all sides of the leaf until water is squeezed from the petiole. The greater the pressure within the chamber, the more water stressed the leaf is as it must be pushed incredibly hard to secrete water. This measurement can be used relatively universally across plants to establish a stress level. The predawn water potentials are the control because they are taken before the plant has begun photosynthesizing and thus contain the most water, however across both middays and predawn you will see a seasonal decrease in potential as the leaves become drier. These measurements can also be compared year to year to assess drought severity.

Remote sensing can also be used to identify landscape level drought stress. Aircraft imagery used in Sequoia Kings Canyon National Park was used to analyze canopy water content as well as live and dead status of tree crowns. Live/dead status can be taken by simply observing the crown of Sequoias and judging color. Any green within the foliage would categorize a tree as live, however if a majority is brown we would assume death

within the following years. If a tree crown is fully brown we would label it as dead. CWC is broadly correlated to leaf water content and serves as an indicator of progressive drought-related stress[3]. High fidelity imaging spectroscopy (HiFIS) was used to collect data on CWC within Giant Sequoia groves. HiFIS takes high resolution spectral images to assess the absorption and scattering of light in the electromagnetic spectrum relative to atmospheric water and water within the canopy (Green, 2003)[7]. Essentially CWC is the total amount of water within the canopy.These images can also be put into digital models to create maps of highly stressed areas.

Drought:

Drought stress is something that most plants across California face at one time or another. Due to the Mediterranean climate mild, wet winters are contrasted by long, hot, dry summers. This extreme climatic shift is one reason why California hosts such a diversity of species as plants must be specially adapted to deal with low water availability throughout an approximate 6 month summer period. However between 2012 - 2016 California faced its most extreme drought in over 120 years. While we were letting our front lawns die and popularizing the idea that our hair didn't really have to be washed on a daily basis there were much more fundamental changes happening. This stress caused responses in Giant Sequoias from the leaf all the way to the landscape level. Water potentials from tree top samples indicated that sequoias approached or exceeded several physiological thresholds, including 50% loss in hydraulic conductivity, leaf turgor loss point (the point at which a leaf begins to wilt), and air entry into the xylem causing a loss of conductivity effectively preventing the branch and leaves from receiving water[3]. While water potential measurements did not show a strong correlation to CWC, they are a standalone indicator of a tree's first response to stress.

CWC was also found to be a sound predictor of changes at the canopy level. Trees at lower elevations and further from water sources showed a greater loss in CWC. Shorter trees also showed lower CWC which contradicts the belief that larger trees are more susceptible to drought stress. The Findings from Nydick's study indicate that sequoias undergoing drought stress begin by making leaf level adjustments, and as conditions become more severe they transition to partial canopy foliage dieback to reduce CWC. While overall mortality during extreme drought was low it could make Giant Sequoias more susceptible to biotic and abiotic threats, namely fire and bark beetles.

Bark beetles, while previously thought to primarily be a threat to pine species in mixed conifer forests, are now hypothesized to be an increasing mortality agent in giant sequoias. Until recently they have only been found in dead branches or fallen limbs of giant sequoias but under severe water stress their defenses are weakened potentially making them more vulnerable to attack. While their exact role in disease and mortality within Giant sequoias requires more research, their harm to dead wood within a live tree can increase vulnerability, especially when mining around a fire scar[4]. Bark beetles and other insects are just one of the biotic factors that can compound stress among droughted trees. There are also abiotic factors such as fire which can greatly increase the mortality rate during drought.

While past studies did not conclusively show that fire scars increase mortality among mature sequoias, the current research posits that under drought stress, previously non-lethal factors, such as fire damage, could become more significant and lead to mortality.

Fire:

Fire, while a natural process and one necessary to maintain the landscape of mixed conifer forests has become increasingly man-made causing unnatural frequency, intensity and severity in recent history that poses a threat to ancient trees including the Giant Sequoia. In the last 50 years the majority of wildfires are human caused. From baby showers to poorly maintained power lines humans have managed to wreak havoc on the landscape. The natural burn regime for conifer forests is between 1-36 years. While the frequency might fall within this spectrum the intensity is well outside as the natural burn size is typically around 100 Ha, however in the last century they have ranged from a couple thousand to a couple hundred thousand acres. In California in June 2021 the Dixie fire burned more than 463,000 acres making it the second-largest in California's history. While tragic and incredibly destructive, California has seen a number of devastating wildfires quite recently. The Campfire, the Tubbs fire, and the Oakland Hills fire, all occurred within 30 years of each other.

Sequoias in particular possess serotinous cones which must reach a certain temperature before releasing their seeds. This helps to remove competition in what can often be a dense forest in which seedlings must fight for light and nutrients. Fire also has the capability of resetting the soil and removing organic build up. One may initially think that such a well adapted tree, especially one with a lifespan of nearly 5,000 years would not be phased by an increase in fire, as it’s longevity is built upon its resistance to pests, disease and disturbance, but evidence suggests that while initial mortality due to burn appears low there are a number of confounding factors that increase the threat wildfire poses.

With the unnatural buildup of fuel throughout the 20th century fire suppression movement the increase of fire threatens the vast ecosystem benefits that ancient forests provide including carbon storage, biodiversity, hydraulic and watershed functions. When trees are removed from a landscape evapotranspiration is halted and large root systems no longer hold soils in place. This can cause massive landslides and change the chemistry of watersheds as huge increases in sediment enter streams and rivers. While a complete lack of fire does not allow for regeneration of the Giant sequoia as its cones are not able to drop seed, too much can lead to continual addition of scars which have been found to increase mortality in years post fire. Mortality within groves is often assessed via remote sensing in a similar manner to canopy water content in which trees are observed from

an aerial view to track crown damage. It was found that trees with greater percent crown damage were more susceptible to mortality 1-5 years after a fire, and trees were 2.4 times more likely to die if they possessed a fire scar from a previous burn[2]. In the case of giant sequoias, burn scars can be thought of as death by a thousand cuts. Dendrochronologists can use tree cores to find centuries worth of fire data as scars leave permanent marks in the wood of dicots that can show both the severity and range of a burn. Once a tree forms a scar that portion of the wood permanently dies, however surrounding wood can grow around the scar, which is what creates a permanent mark in the core. Mortality becomes a factor when the cambium does not have adequate time to regenerate in between highly intense fires.

Cambium damage in combination with other factors such as drought and disease (bark beetles) can lead to elevated mortality. Anthropogenic agents of climate change and ecosystem manipulation lead to compound effects on some of the earth's oldest and most valuable resources. Even highly adapted plants are no match to our impact and when Giant sequoias are vulnerable from fire scars intense drought further leads to foliage die back. This weakness also allows pest entry that together can take down 5,000 years of survival. Knowing which metrics best quantify damage and lead to mortality can in turn help us create management techniques to counteract adverse effects. Management is key to preserving what little distribution these trees have left and making sure that biodiversity can persist.

Management:

Management is a critical part of ensuring the stressors threatening Giant Sequoias and ancient forests as a whole do not bring about their demise. Having started this problem in the first place it is our responsibility to the trees to protect what little is left. Since 1750 approximately 600 plant species have gone extinct, with many more on their way out. We have an opportunity through effective management to see that fire, drought, and disease do not become insurmountable obstacles. This is also why it is important to identify the biggest threats, along with past strategies that have been used to assess next steps. A relatively recent technique that has arisen is ecosystem management. Ecosystem management is a holistic approach to resource management which accounts for the health of the ecosystem while balancing human and societal needs, such as economic value and the urban wildland interface. It is a step towards thinking about the forest as a whole and the plethora of livelihoods it affects.

In the past management has been less than ideal. Clearly Donald Currey proved that there was something to be desired in terms of regulations surrounding scientific research and one's ability to work in ancient forests. In 1964 the same year that Currey accidentally killed the world's oldest tree the US Forest service began experimenting with prescribed burns in sequoia national park to remove organic buildup. Then in 1985 the first symposium for giant sequoia management was created and attended by scientists and interest groups. In 1990 a professional mediator was hired to resolve conflict within said interest groups, in 1992

president Bush signed a proclamation endorsing the protection of Giant Sequoias from commercial activity, in 1996 a proposal to remove hazardous trees was brought about, and the following year it was shut down, and in 2001 ecosystem management came into play[5]. While it seems as though much discussion has taken place over the last 75 years there are very little coordinated efforts across the federal, state, and regional levels, president Bush's proclamation marking one of the first.

Imagine that you are sitting at the table at thanksgiving dinner, with not just your parents and siblings but the whole kitten caboodle of aunts, a drunk uncle, a deaf grandparent and a few angsty teenagers sprinkled in. While you're sitting in the corner next to your dad you are trying to yell across to aunt janice to pass the cranberry sauce. But aunt janice can't hear you, so instead your weird cousin joey hands you a roll, which isn't even what you asked for in the first place. This is essentially the level of convolution that goes on trying to get federal policy passed to protect a species like the Giant sequoia. While everyone may care in theory there are so many voices at the table shouting across for what they want it's hard to decipher any real words at all, and even harder to get something through to the top. Then all the sudden your uncle clinks his glass, stands up, and says that he really doesn't think we need to do anything at all because if the trees have made it this long he's sure they'll be fine for another couple thousand years. But what he doesn't understand is that if any of us are to make it even 100 years farther into the future action needs to be taken right now. Like right now right now. So by assessing how damage like fire scars, and extended droughts lead to a higher vulnerability we can best decide how to pinpoint our action.

In terms of drought we can reduce outside stressors such as infrastructure that alters soil, such as water extraction facilities so that as much water as possible can be given to the trees[3]. For fire, forests can be mechanically thinned and treated with prescribed burns to manage fuel loads that would otherwise lead to highly severe burns. For areas which have already received beneficial fire (low severity) they are in good condition for priority retreatment with prescribed fire to maintain a natural regime[2]. Given our current workforce, funding, and compliance metrics we are still beyond landscape scale restoration. It is vital that federal, state, and local agencies find a way to communicate. Interest groups such as non-profit organizations are key proponents of action and are a great way for the public to get involved in protection.While it may sometimes feel like futile action to fight for protection of what we believe in, it is never a waste. We shall not witness the end of giants, they were here before us and we will ensure that they are here after us.

References:

1. Bristlecone Pines, The Prometheus Story. National Park Service, Great Basin National Park Nevada. [Updated August 12, 2023]. https://www.nps.gov/grba/planyourvisit/identifying-bristlecone-pines.htm#CP_JUMP_63561

2. Shive, K. L., Wuenschel, A., Hardlund, L. J., Morris, S., Meyer, M. D., & Hood, S. M. Ancient trees and modern wildfires: Declining resilience to wildfire in the highly fire-adapted giant sequoia. For. Ecol. Manage. 2022; *511*, 120110. https://doi.org/10.1016/j.foreco.2022.120110

3. Nydick, K. R., Stephenson, N. L., Ambrose, A. R., Asner, G. P., Baxter, W. L., Das, A. J.Dawson, T., Martin, R. E., & Paz-Kagan, T. Leaf to landscape responses of giant sequoia to hotter drought: An introduction and synthesis for the special section. For. Ecol. Manage. 2018;*419–420*, 249–256. https://doi.org/10.1016/j.foreco.2018.03.028

4. Douglas D. Piirto. Giant Sequoia Insect, Disease, and Ecosystem Interactions. USDA Forest Service Gen. Tech. Rep. 1994; PSW-151.pages 82-89. https://digitalcommons.calpoly.edu/cgi/viewcontent.cgi?article=1012&context=nrm_fac

5. Rigg, C. M. Orchestrating Ecosystem Management: Challenges and Lessons from Sequoia National Forest. Conserv Biol. 2001; *15*(1), 78–90. https://doi.org/10.1111/j.1523-1739.2001.99339.x

6. OpenStax Biology [Image credit]. Organismal Biology. [accessed 2025 May 18]. https://organismalbio.biosci.gatech.edu/nutrition-transport-and-homeostasis/plant-transport-processes-i/
7. Green RO. Understanding the three phases of water with imaging spectroscopy in the solar reflected energy spectrum. United States -- California: University of California, Santa Barbara; 2003. 219 p.: Last updated - 2023-03-02. Retrieved from https://www.proquest.com/dissertations-theses/understanding-three-phases-water-with-imaging/docview/305341860/se-2